Architecture-Centric Development

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Objectives: Architecture-Centric Development

- Describe the “4+1 view” model of architecture
- Discuss the responsibilities of the architect
- List the contents of the Software Architecture Document
- Define patterns and frameworks
- Explain layered architectures and provide layering guidelines
- Describe how architectural mechanisms are identified and mapped to design and implementation mechanisms

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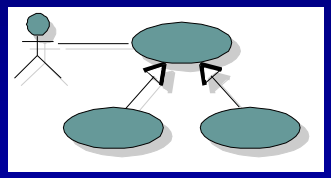
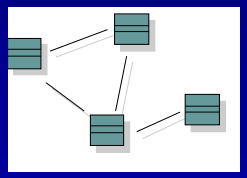
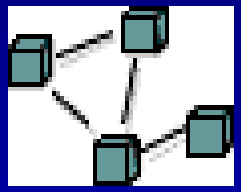
Architecture Is Represented by Views



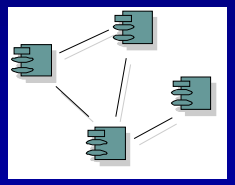
Use-Case Model



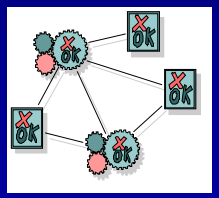
Design Model



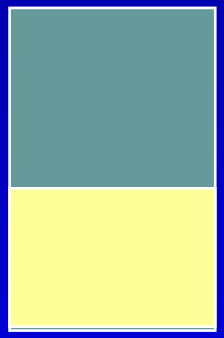
Deployment Model

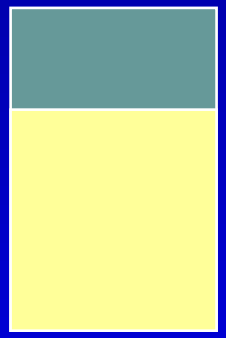


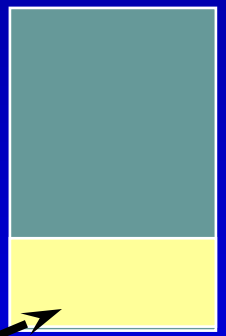
Implementation Model

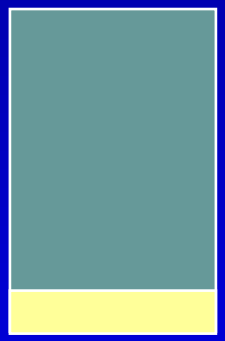


Test Model

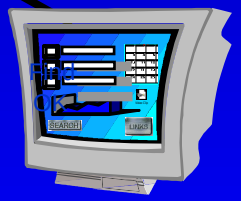


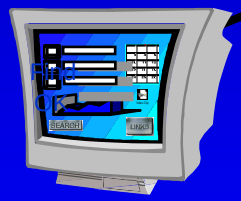






Use Case View





Implementation View

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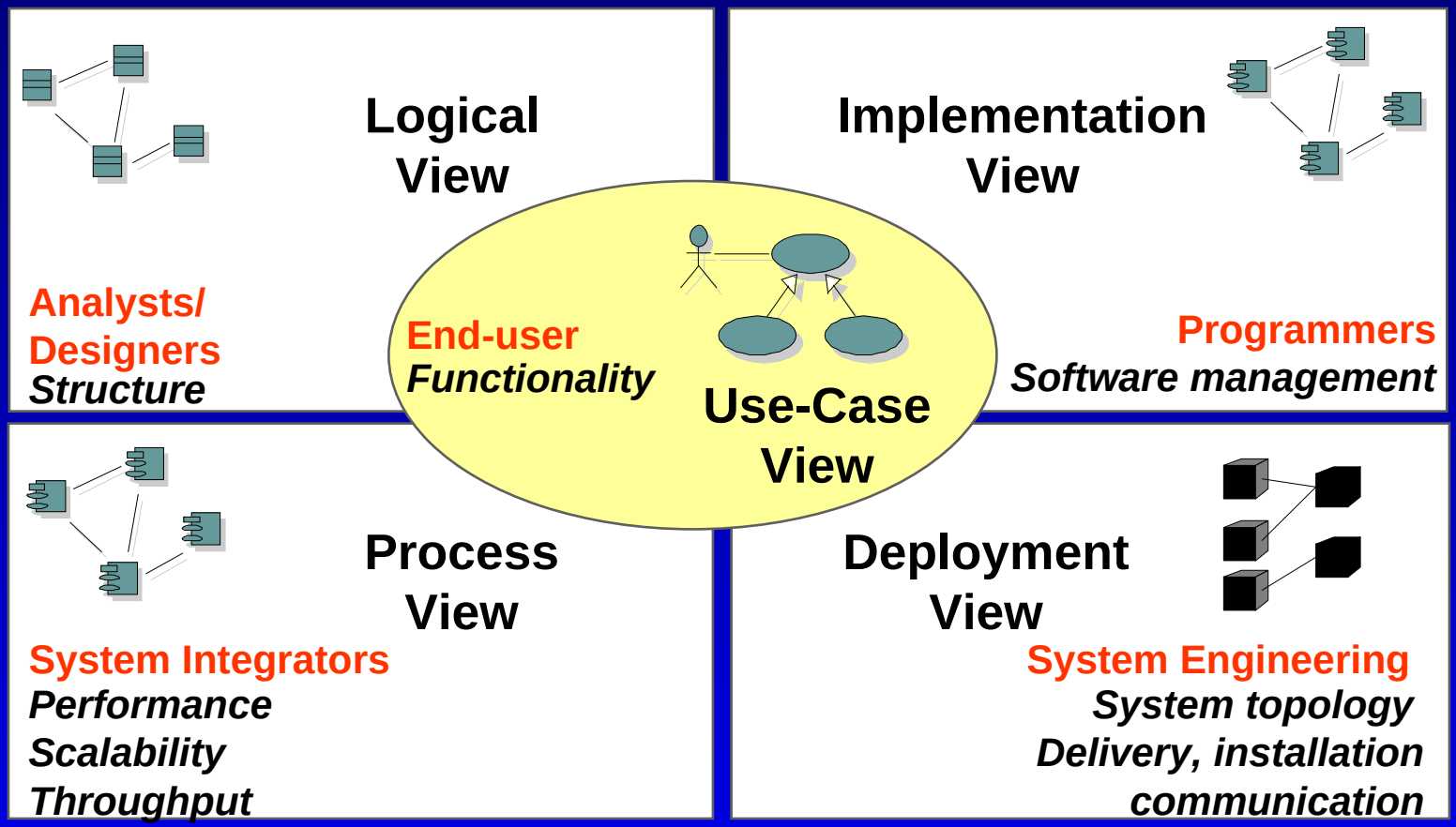
A “4+1 View” Model of Architecture



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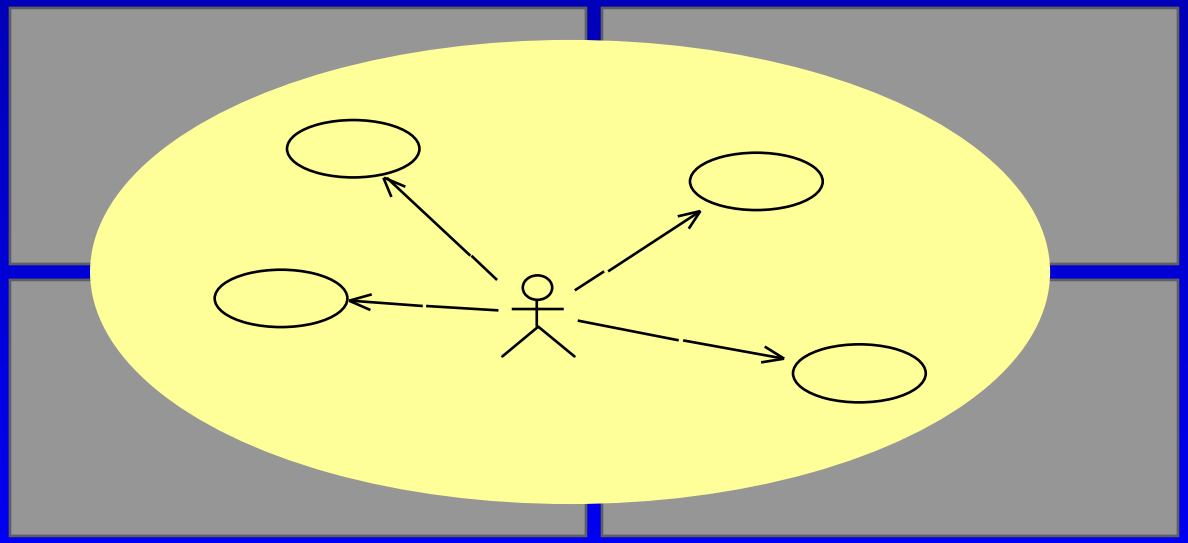
Use Case View

- Describes an architecturally significant subset of the use-case model (i.e. a subset of the use cases and packages)
- Concerns: functionality, critical functions, performance
- Represented, graphically, on Use Case Diagrams



- **Actors**
- **Use Cases**

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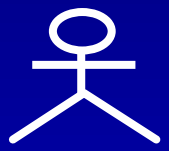
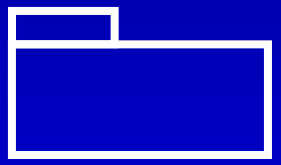
Use Case View Modeling Elements

Components



Actor

Use Case



Package

Connectors

Generalization or Inheritance
<<uses>>
<<extends>>

Communicates

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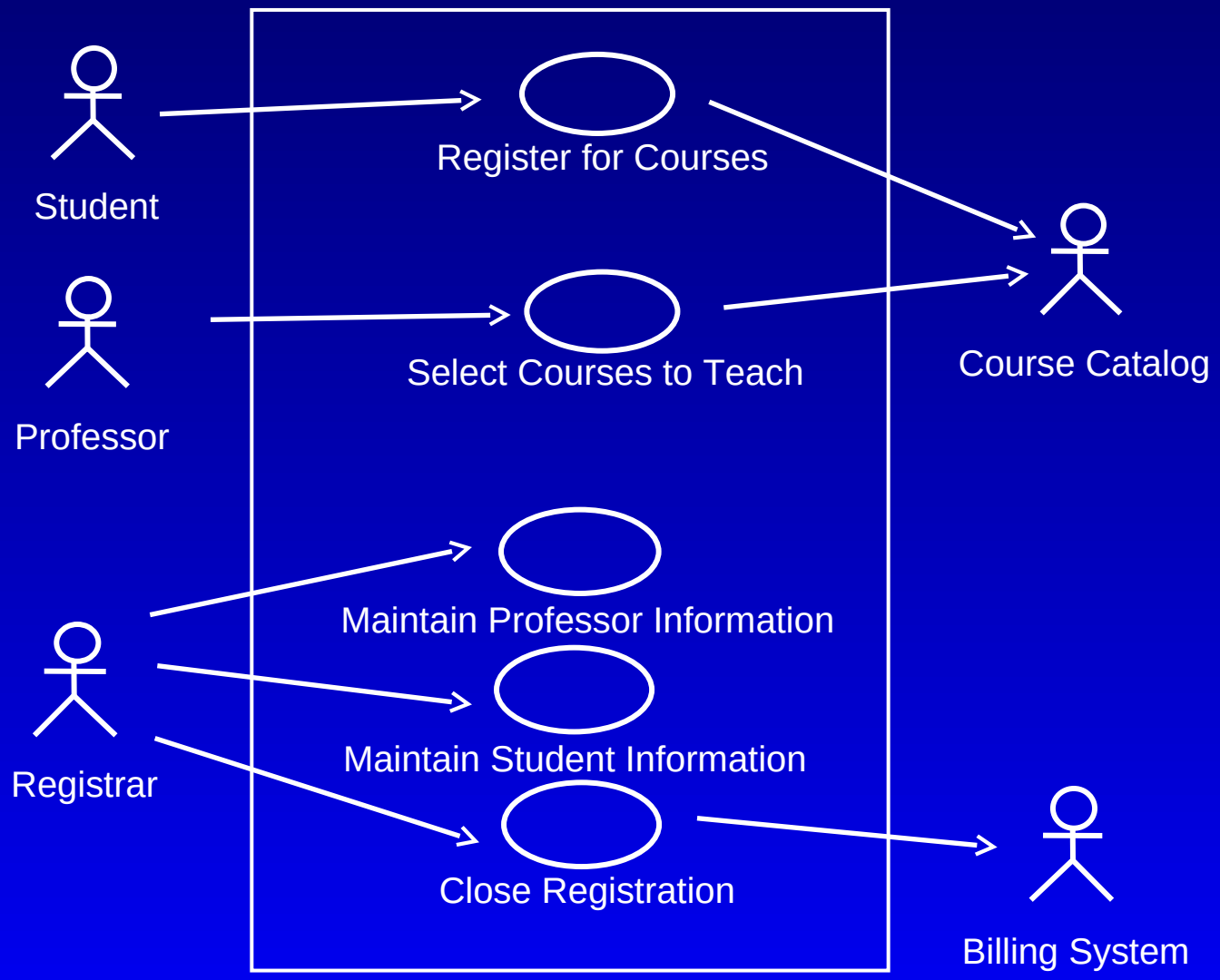
Use Case Diagram: Course Registration System



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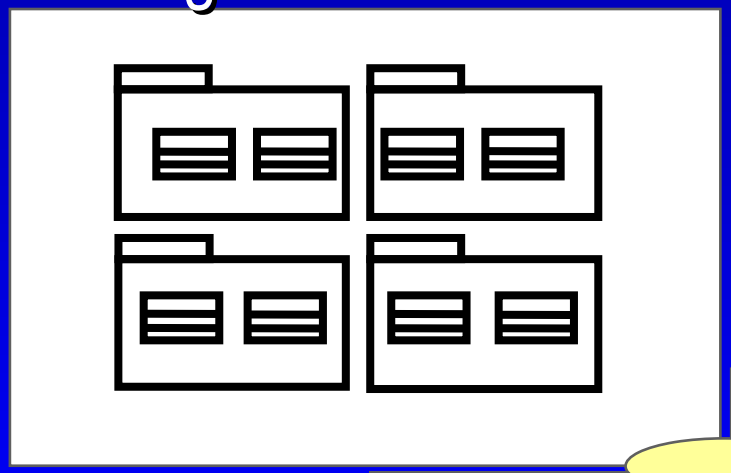
Logical View

- Describes an architecturally significant subset of the design model, i.e. a subset of the classes, packages, and use-case realizations.
- Concerns: functionality, behavior, use of frameworks, patterns
- Represented, graphically, on Class Diagrams, Interaction Diagrams and State Diagrams



- **Package structure**
- **Interesting classes**
- **Use Case Realizations**

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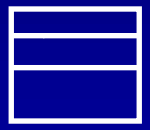
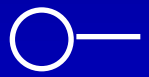
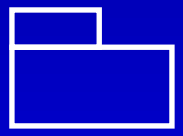
Logical View Modeling Elements

Components


class

state


interface


package

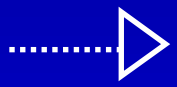
object

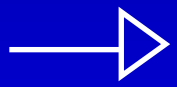
use case realization

Connectors

Association or link or transition

aggregation


realization


generalization

dependency

message

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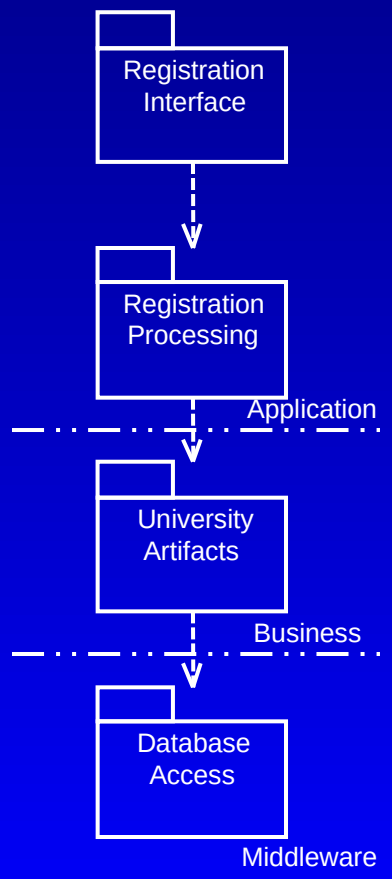
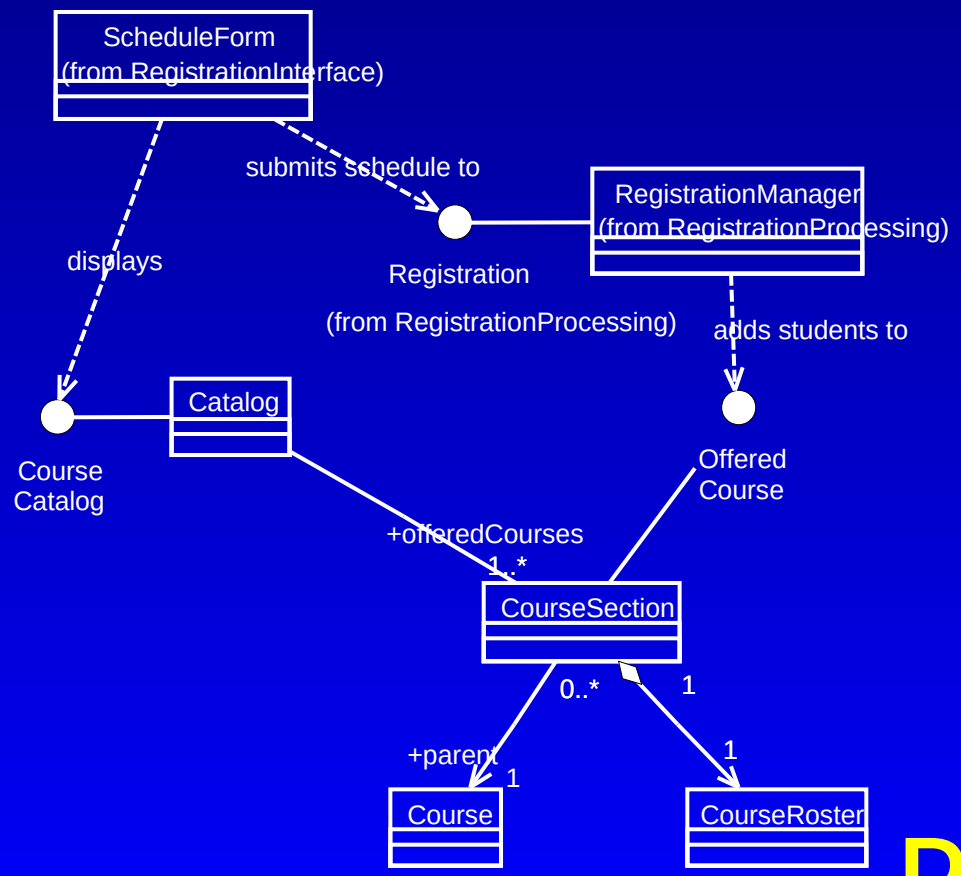
Class Diagram

- Registration System

Main Class Diagram



UniversityArtifacts Main Class Diagram



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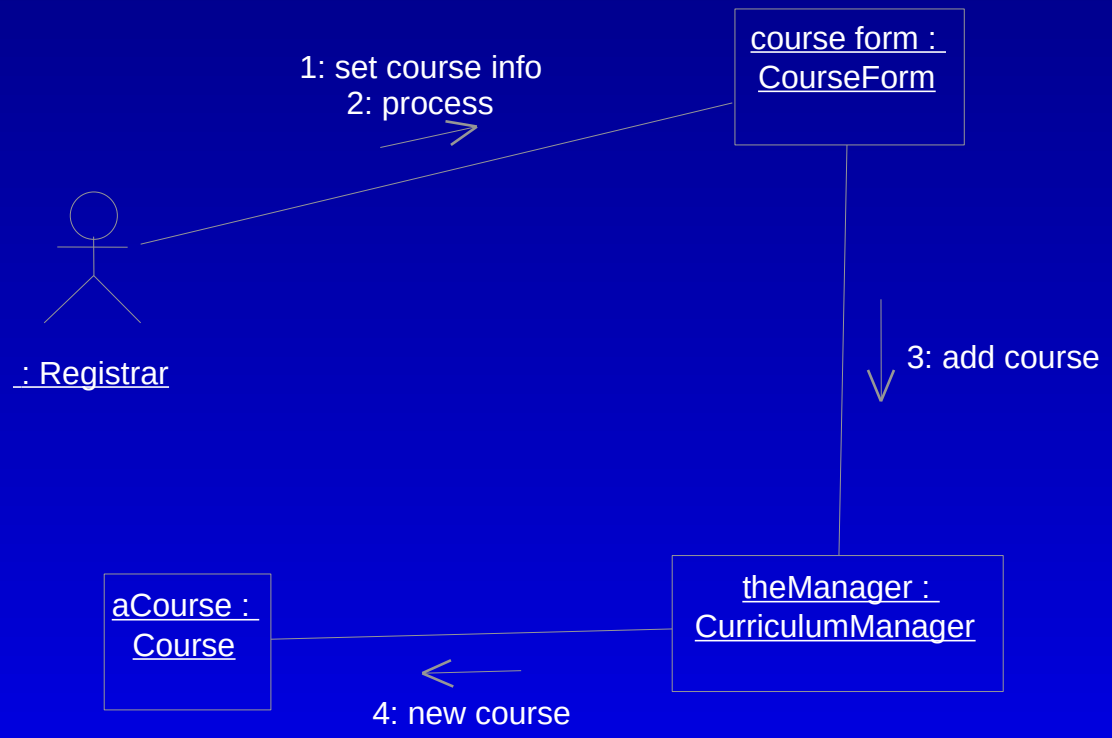
Collaboration Diagram

- Maintain Course Info Use Case (partial)



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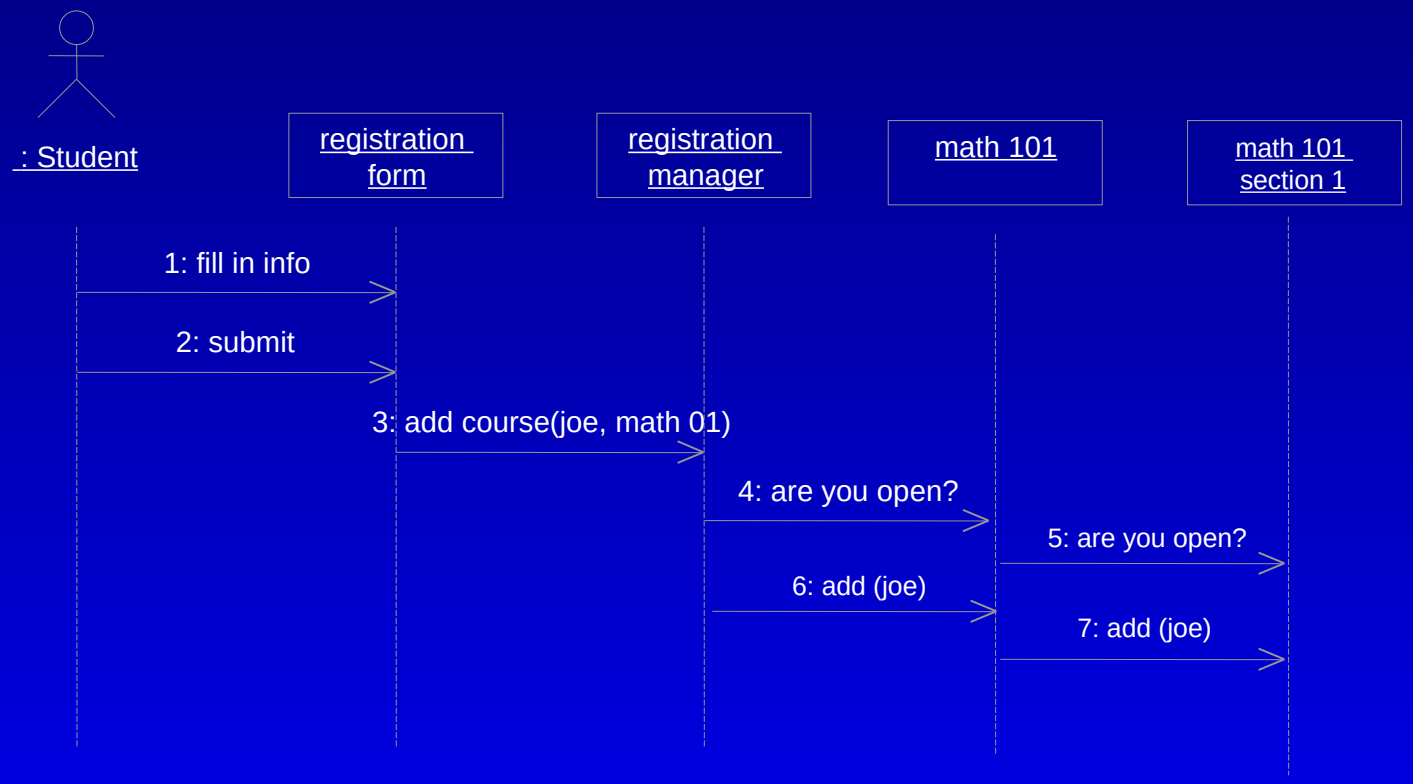
Sequence Diagram

- Register for Courses Use Case (partial)



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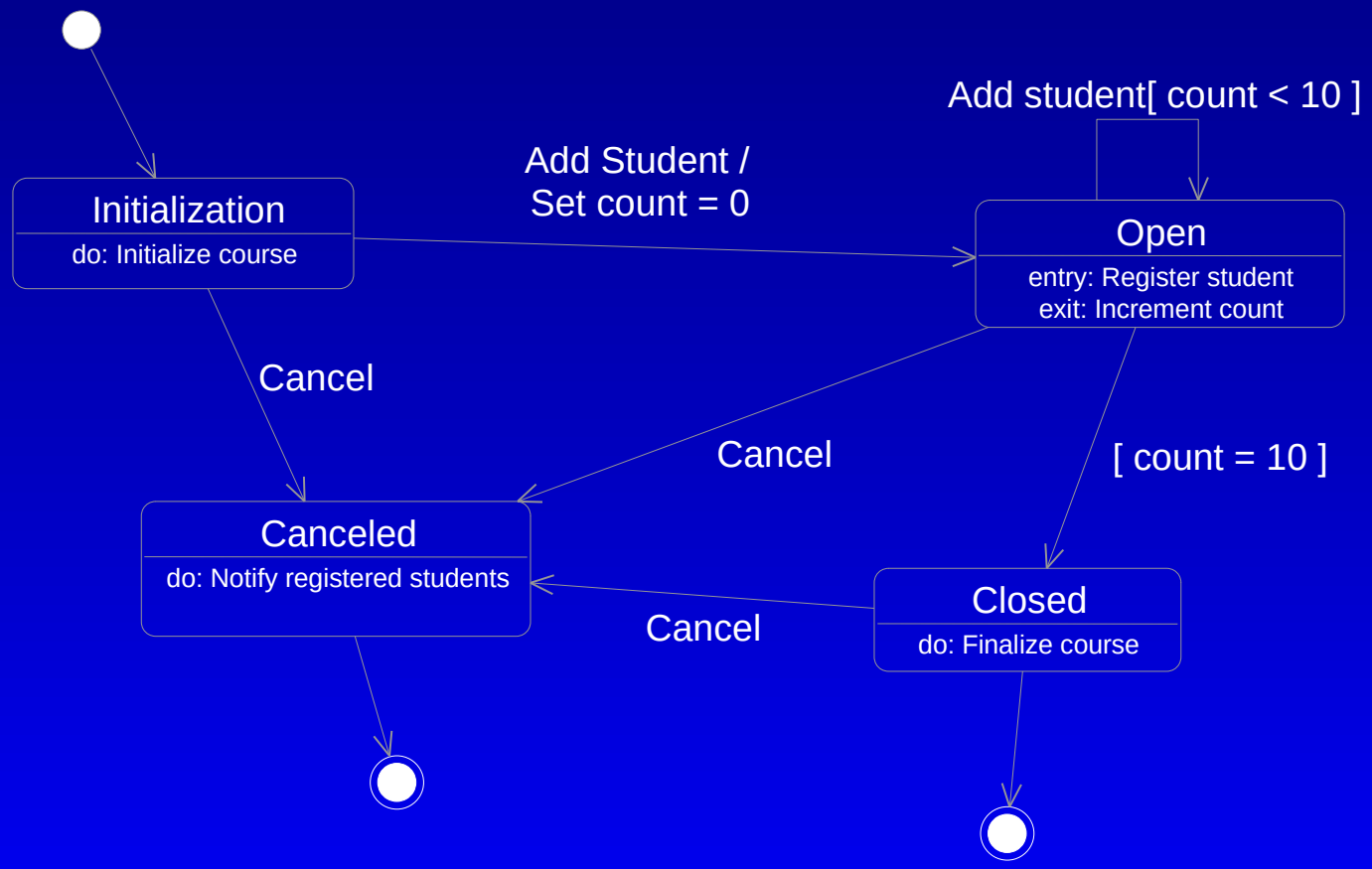
State Diagram

- State Diagram for the Course class



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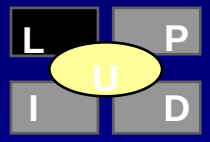
From Use Case View to Logical View



- Find Classes from Use Case Flow of Events
- Allocate the use case responsibilities amongst the identified classes
- Develop the use case realizations (e.g., “visually model” the use case)
- Identify the attributes and relationships needed to support the use cases
- “Package” the identified classes, refining package relationships, as necessary

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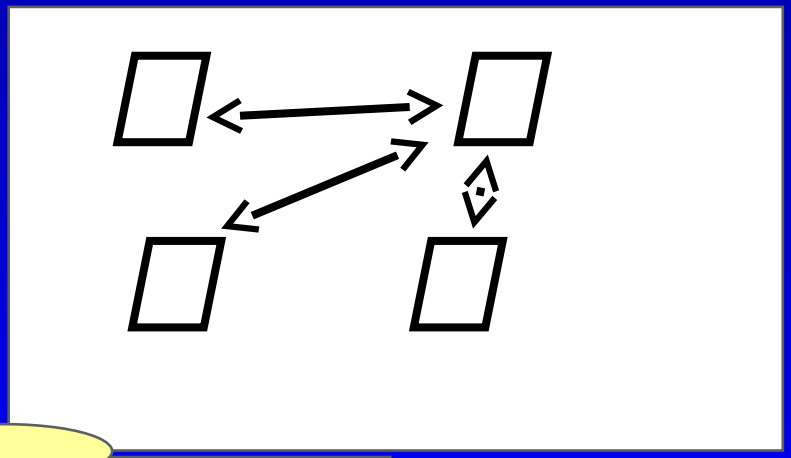
Process View

- Describes the system's decomposition into executing tasks and processes
- Concerns: availability, reliability, scalability, integrity, performance, system management, synchronization
- Represented graphically on Component Diagrams and Class Diagrams

•Tasks
•Processes
•Process structure



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Process View Modeling Elements

Component

Process1.exe

Process or Task (Active Object)

«stereotype»
Comp Name

<<process>> or <<task>>

interface

Connector

- - - - ->

Dependency(component diagram)

Link (collaboration diagram)

Message(Simple, Synch, Asynch, Timeout, Balking)

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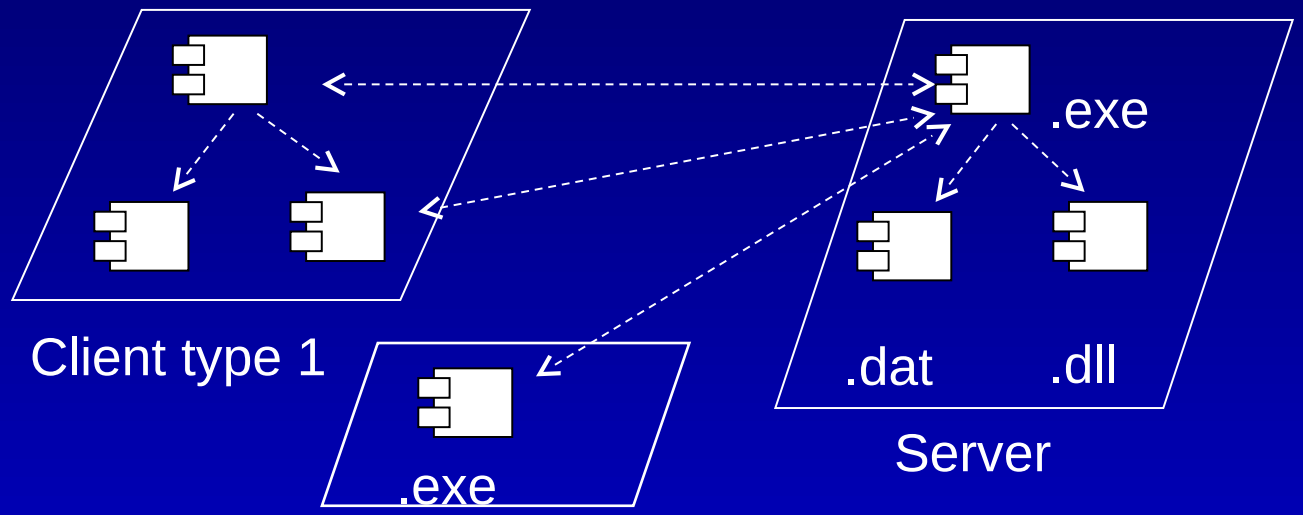
Process View Diagrams

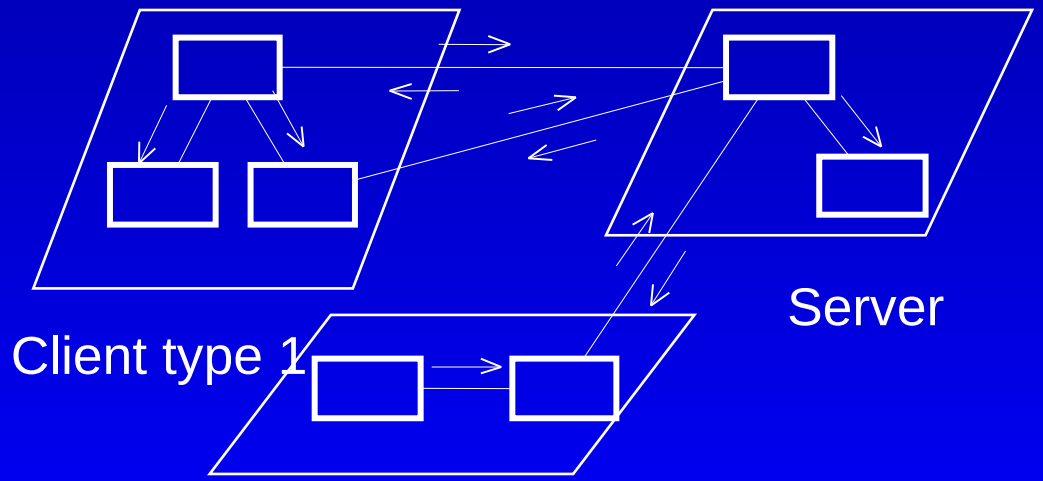
With Components

Component Diagram



With Active Objects

Collaboration Diagram



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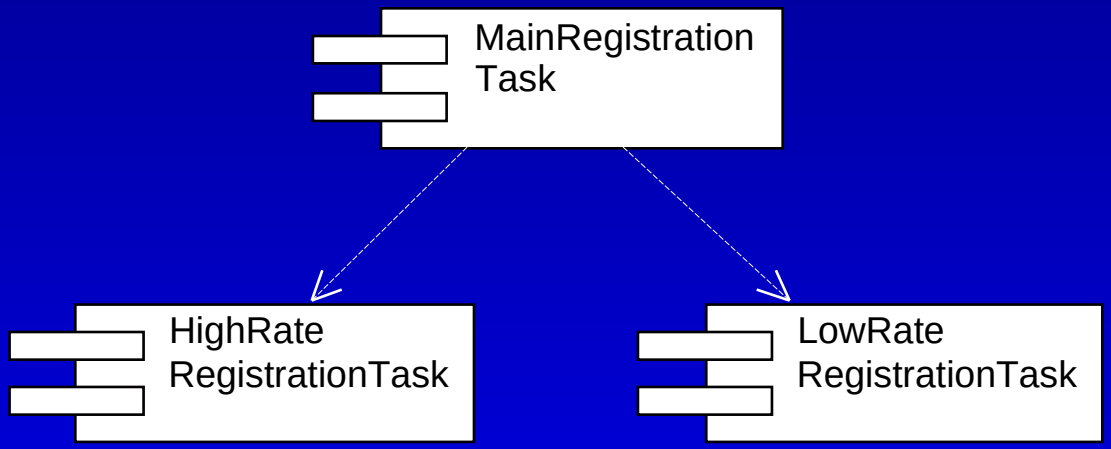
Component Diagram

- Registration Manager Process



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From Logical View to Process View

- A process for each active object is not possible
- Allocate active objects to processes based on:
  - Performance and concurrency requirements
  - Distribution requirements and support for parallel execution
  - Redundancy and availability requirements

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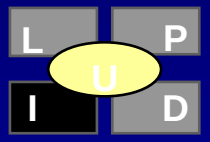
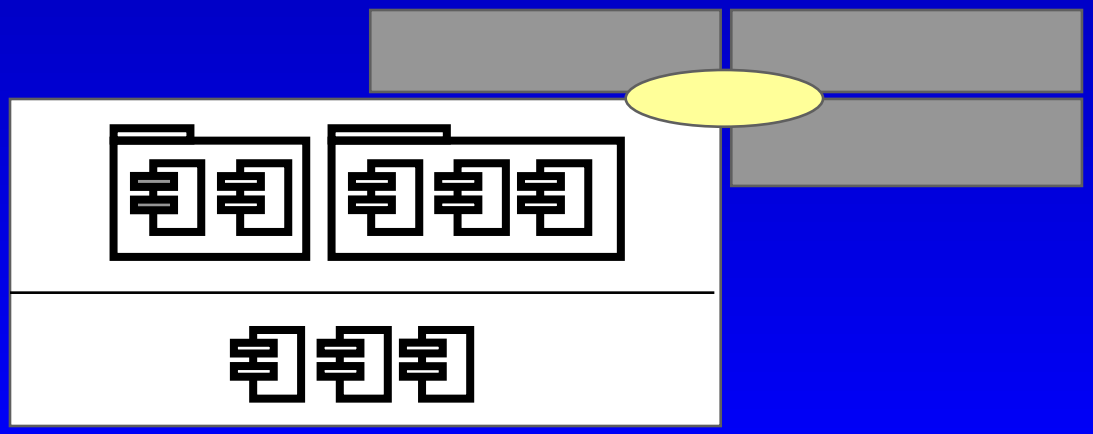
Implementation View



- Describes the software component organization in the development environment
- Concerns: ease of development, team organization, inclusion of existing legacy systems or components, software configuration and management
- Represented graphically on Component Diagrams



•Code library structure
•Source code

Deliverables
.exe
.DLL
Data files

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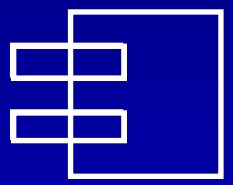
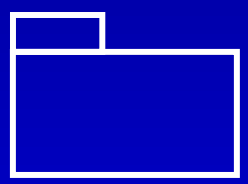
Notation for the Implementation View

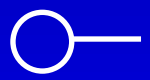
Components



Component
(such as .h, .cpp,
.dll, .exe,
.dat, etc.)



Package



interface

Connectors

.....>

Dependency
Compilation Dependency
(such as #include, “with”,
import, etc)

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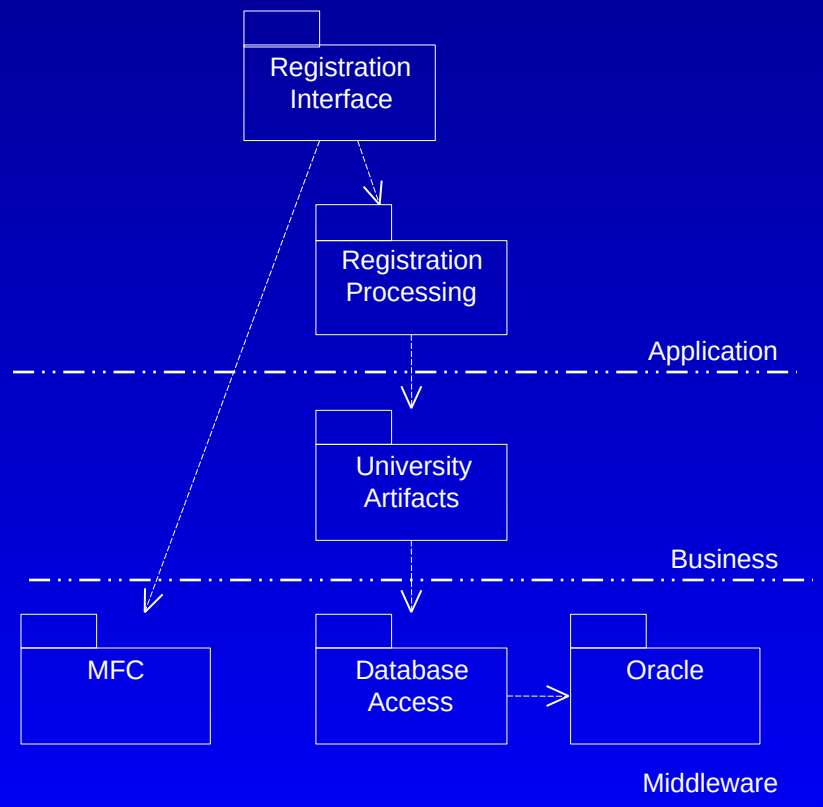
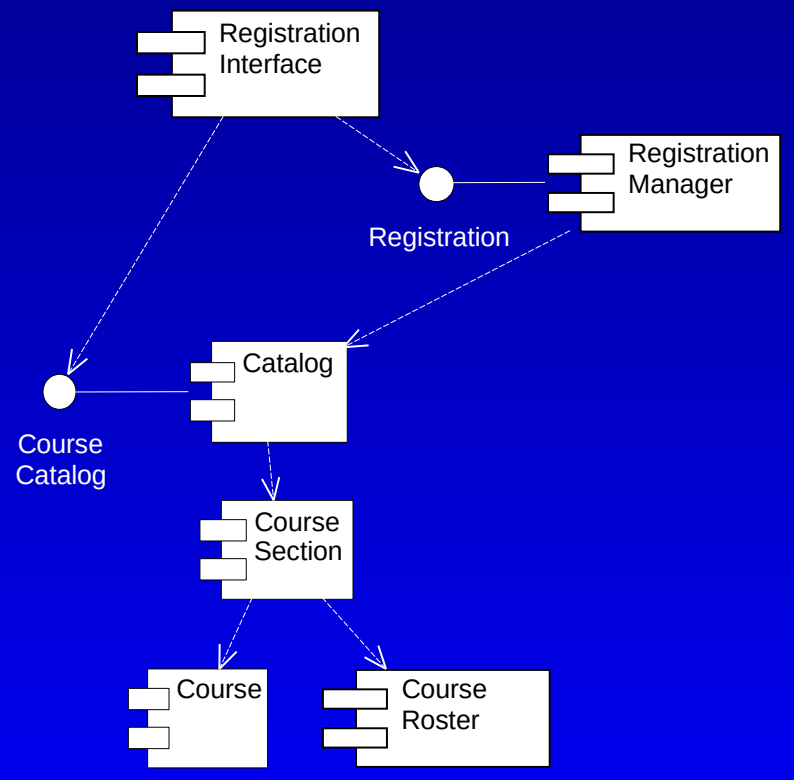
Component Diagram

- Registration System

Main Component Diagram



UniversityArtifacts Main Component Diagram



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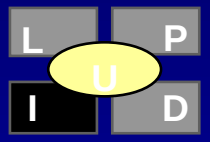
From Logical View to Implementation View



- Map Logical View elements to Implementation View elements
- Considerations:
  - Programming language
  - Team organization
  - Degree of expected reuse and commonality
  - Existing legacy systems or components
  - Strict layering principles
  - Release strategy/policy
  - Configuration Management

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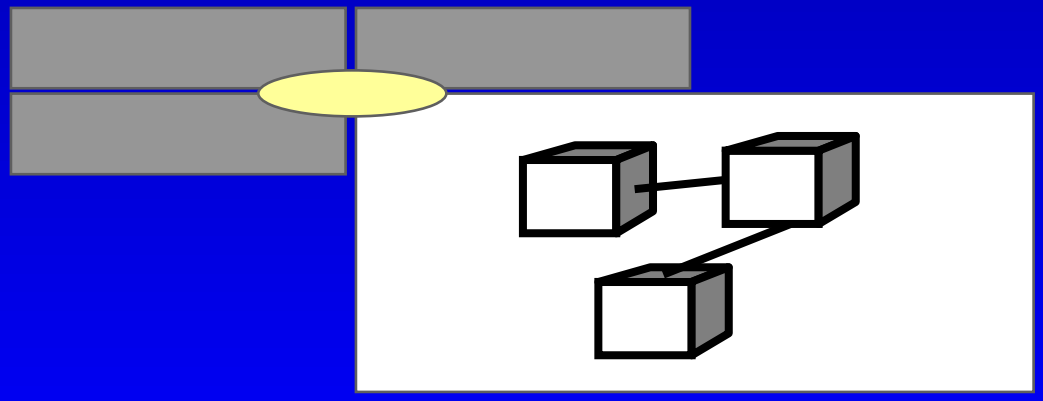
Deployment View

- Describes physical network configurations
  - Physical nodes and their interconnections
  - A mapping of the processes in the process view onto the physical nodes
- Concerns: performance, fault-tolerance, availability, installation, maintenance
- Represented graphically on Deployment Diagrams



Hardware/ Software Mapping

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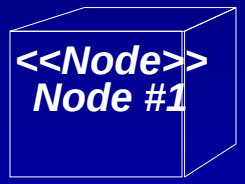
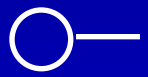
Deployment View Modeling Elements

Components



Node



interface

Runtime component

Process or Task
(Active Object)

Connectors

Communication association

dependency

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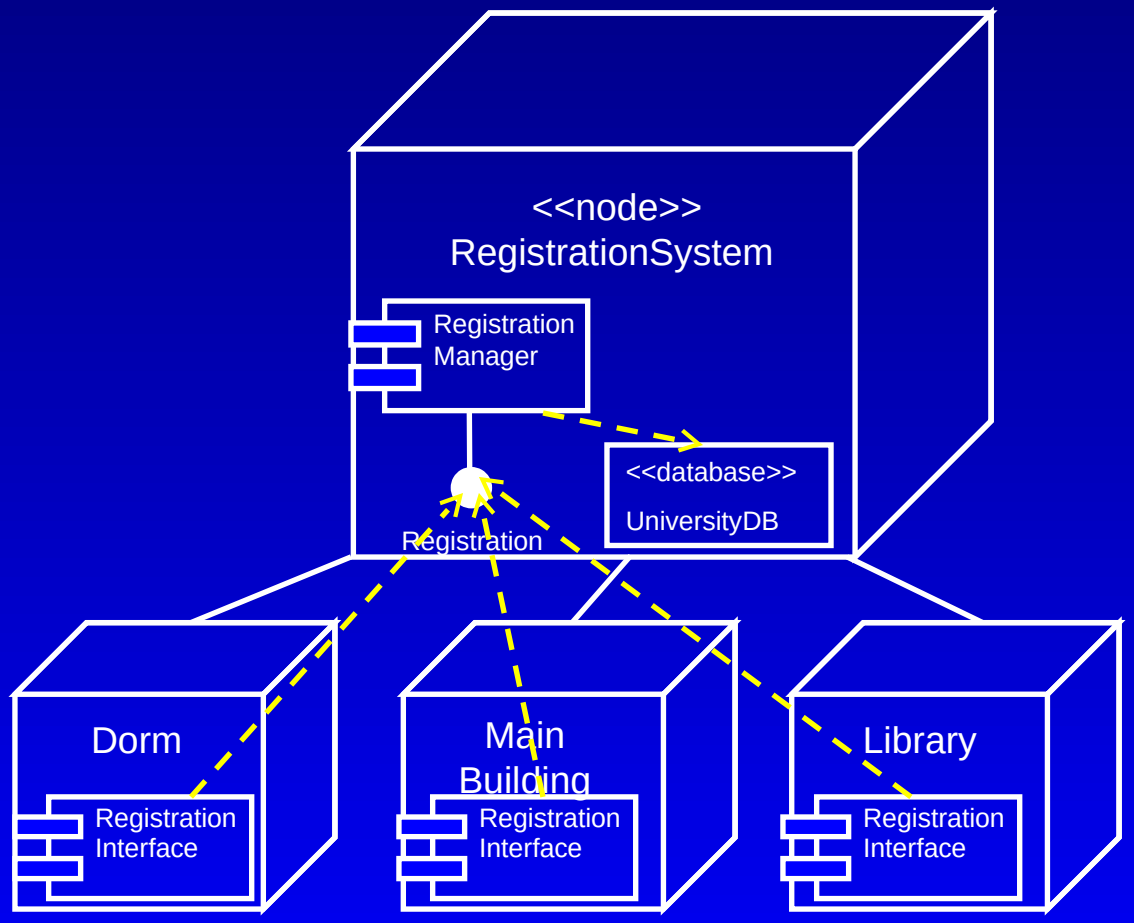
Deployment Diagram

- Registration System



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From Process View to Deployment View

- Map the Process View processes, tasks and/or process groups onto the available physical hardware (nodes) for testing or deployment
- Considerations
  - Node capacity
  - Interconnection bandwidth
  - Availability (Primary/backup)
  - Rerouting
  - Load sharing
  - Parallelism

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Key Worker: The Architect

- Leads and coordinates technical activities, including
  - Architectural analysis and design
  - Determination of system concurrency and distribution
  - Structuring of system models
  - Prioritizing use cases
- Is responsible for the overall content of the system models as well as:
  - Design Guidelines
  - Programming Guidelines
  - Software Architecture Document

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Key Artifact: The Software Architecture Document

- Comprehensive architectural overview of the system:
  - Architectural representation
  - Architectural goals and constraints
  - 4+1 architectural views
  - Data view (optional)
  - Size and performance
  - Quality

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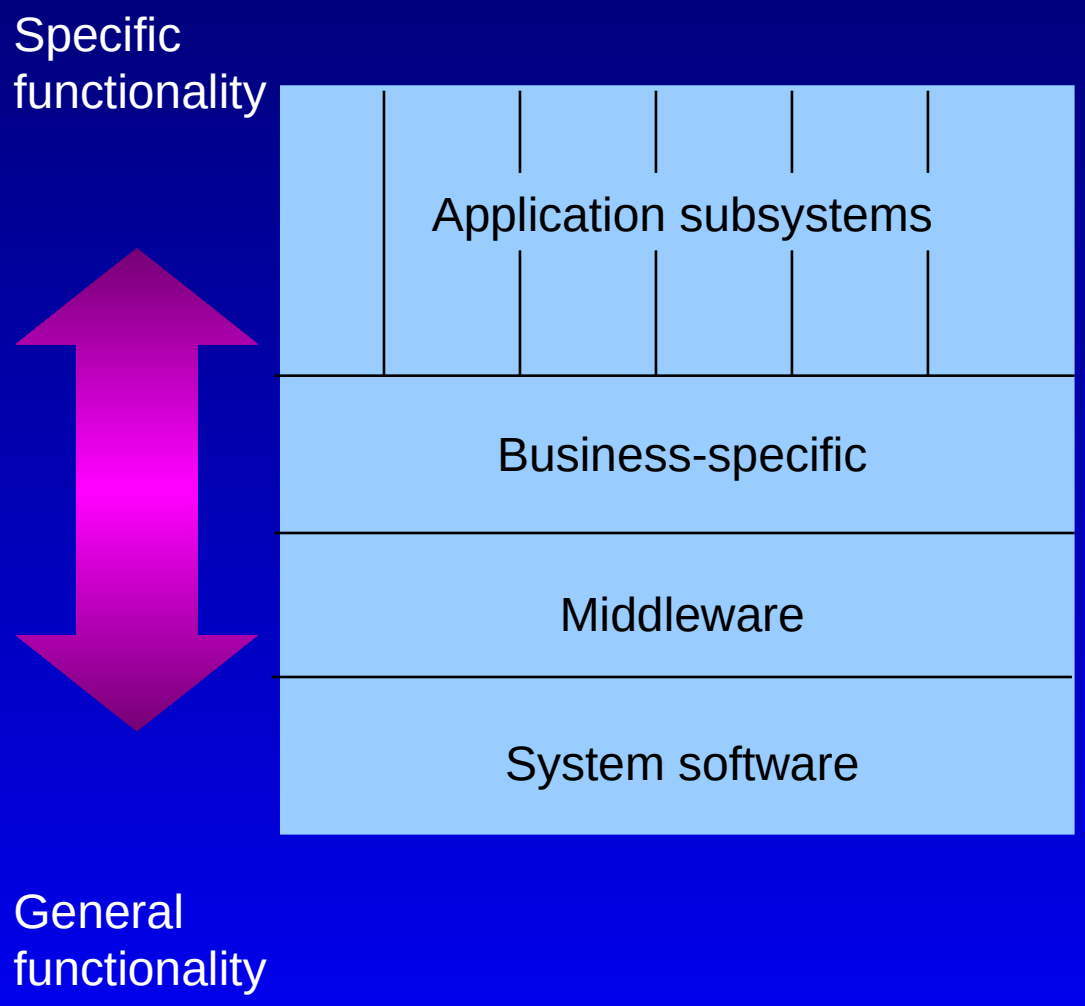
Layers: An Architectural Pattern



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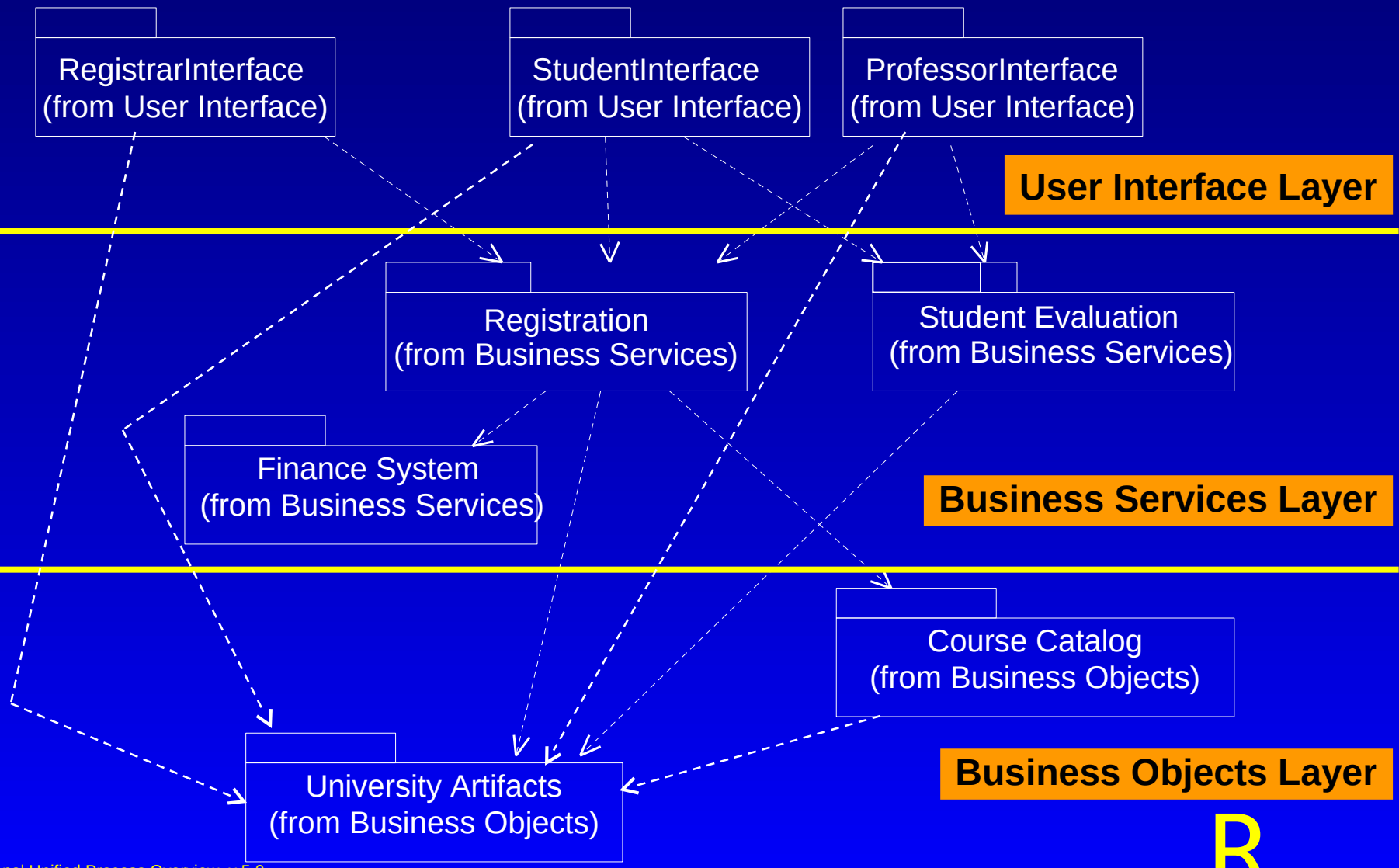
Example: A Layered Architecture



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Layering Guidelines

- Visibility
  - Dependencies only within current layer and directly below
- Volatility
  - Upper layers affected by requirements changes
  - Lower layers affected by environment changes
- Generality
  - More abstract model elements in lower layers
- Number of layers
  - Small system: 3 layers
  - Complex system: 5-7 layers

Goal is to reduce coupling and ease maintenance effort

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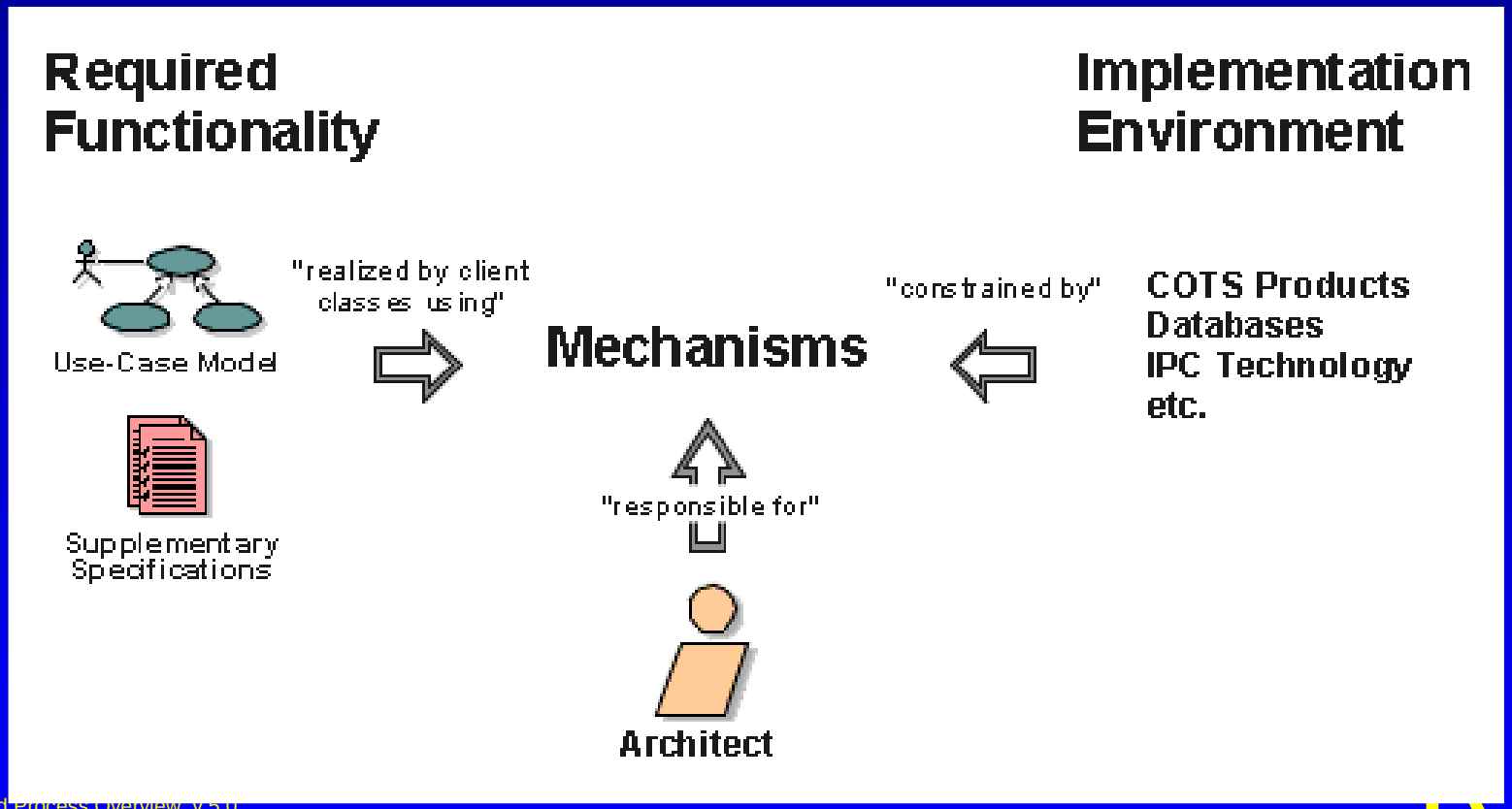
Architectural Mechanisms

- Analysis Mechanisms (conceptual)
- Design Mechanisms (concrete)
- Implementation Mechanisms (actual)



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Sample Analysis Mechanisms

- Persistency
- Communication (IPC and RPC)
- Message routing
- Distribution
- Transaction management
- Process control and synchronization (resource contention)
- Information exchange, format conversion
- Security
- Error detection / handling / reporting
- Redundancy
- Legacy Interface

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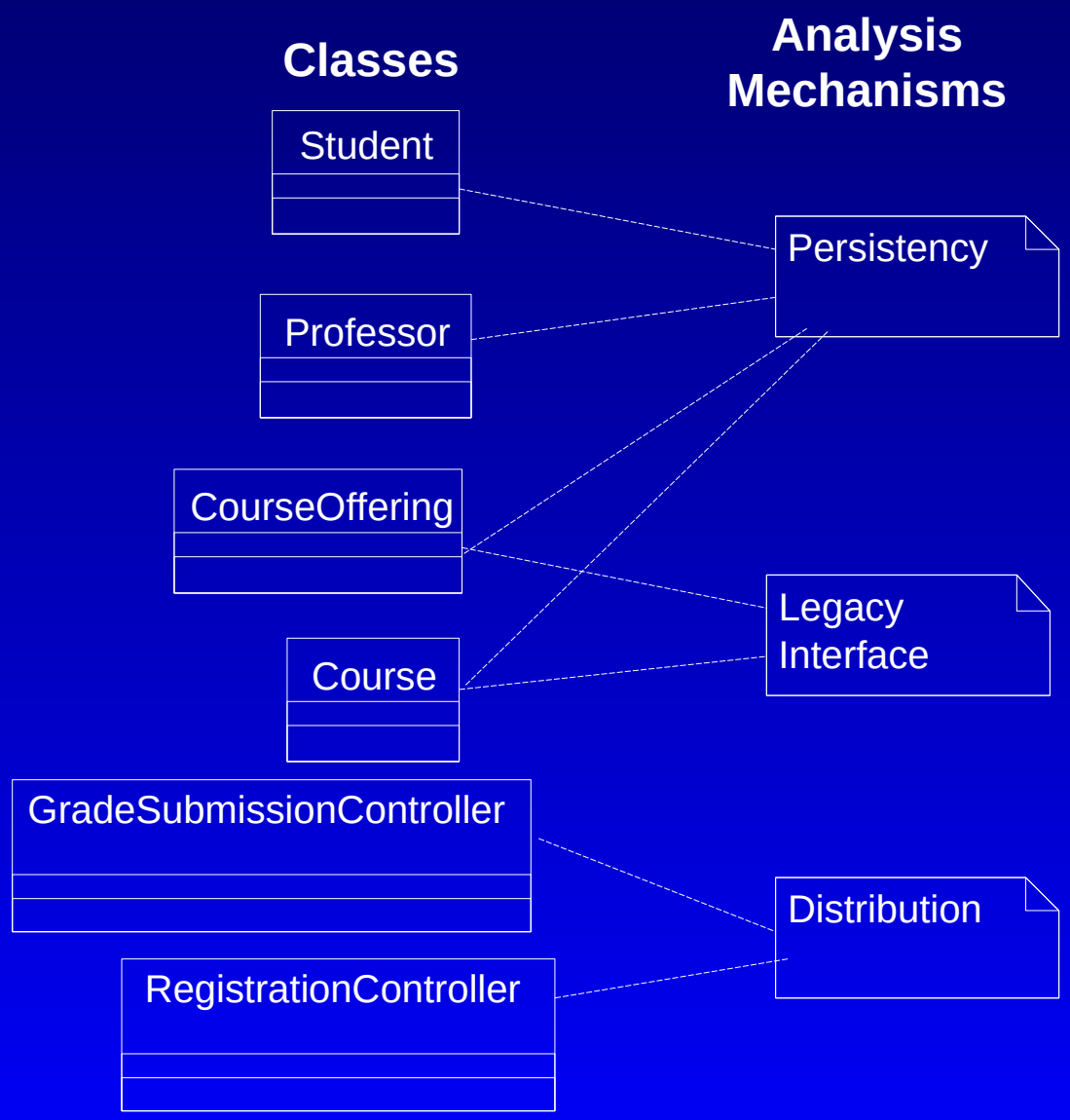
Mapping Classes to Analysis Mechanisms



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Sample Design Mechanisms

- For Persistency
  - In-Memory Storage
  - Flash Card
  - RDBMS
  - ODBMS
- For Communication and Distribution
  - Remote Method Invocation (RMI)
  - Shared memory
  - Function-call-like IPC , Semaphore-based IPC

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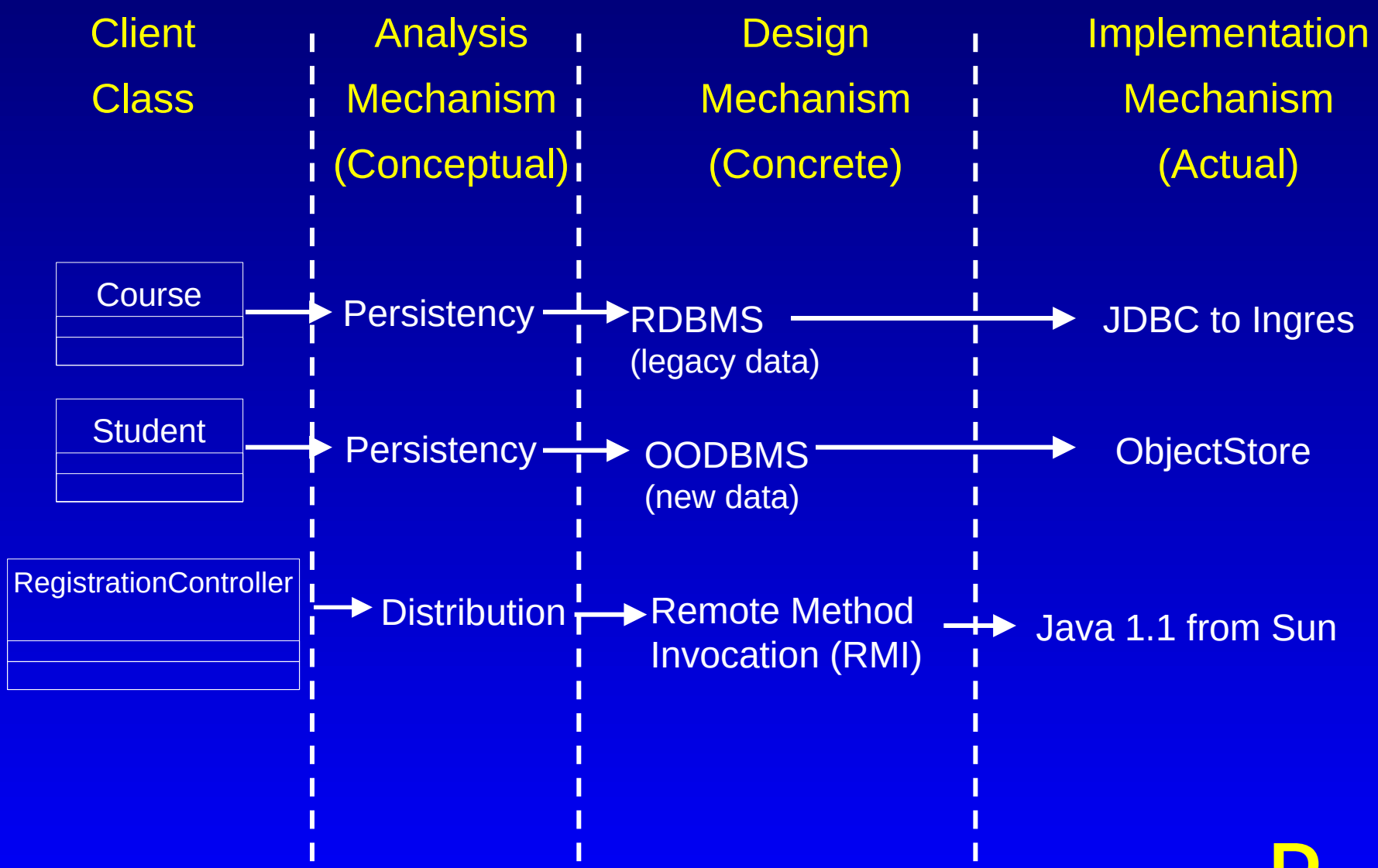
End-to-End Architectural Mechanism Map



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Frameworks and Patterns

- Framework
  - Definition of a general approach to solving the problem
  - Skeletal solution, whose details may be analysis/design patterns
- Pattern
  - A common solution to a common problem in a context
- Analysis/Design Pattern
  - A solution to a narrowly-scoped technical problem
  - A fragment of a solution, or a piece of the puzzle

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Components Are Inseparable from Architecture

- Recall our definition of component:
  - A **component** is a non-trivial, nearly independent, and replaceable part of a system that fulfills a clear function in the context of a well-defined architecture
- Architectural mechanisms can be implemented as components and reused from application to application
- Business services and business objects are commonly implemented as components, intended for reuse
- Component-based development encompasses:
  - Building systems from reusable components
  - Developing and harvesting reusable components

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An Architecture-Centric Process

- Architecture is the focus of the early iterations
- The architect is a key worker
- The Software Architecture Document is a key artifact
- Architectural activities span the requirements, analysis & design, implementation and project management workflows
- Architecture is the foundation for reuse
- Architecture governs design guidelines, including use of patterns and idioms
- Product structure derives from architectural groupings and interfaces
- Team structure also parallels architectural divisions

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Summary: Architecture-Centric Development

- The “4+1” view model of architecture includes the logical, implementation, process, deployment, and use-case views
- The architect is a key contributor on the project, responsible for all aspects of architecture development and documentation
- Patterns and frameworks capture common solutions to common problems
- Layering is a way of organizing an architecture so that it is simple, understandable, and maintainable
- Architectural mechanisms provide common designs and implementations that can be used across a project

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